Ch 3.2: Multiple Linear Regression

Lecture 6 - CMSE 381

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Announcements

Last time:

- 3.1 Linear regression

Announcements:

- Homework #2 Due Sunday on Crowdmark

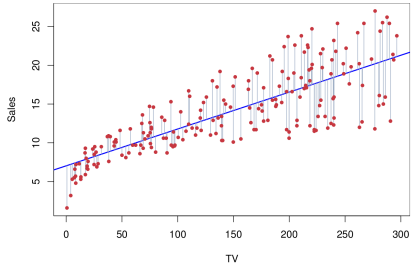
Covered in this lecture

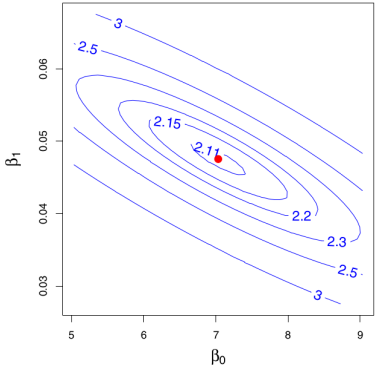
- Multiple linear regression
- Hypothesis test in that case
- Forward and Backward Selection

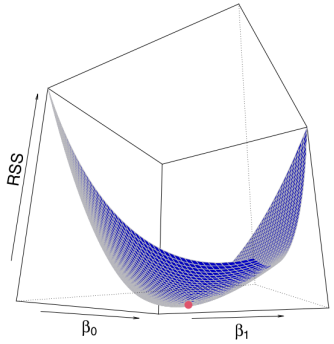
Section 1

Review from last time

Linear Regression with One Variable







- Predict Y on a single variable X

$$Y \approx \beta_0 + \beta_1 X$$

- Find good guesses for $\hat{\beta}_0$, $\hat{\beta}_1$.
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
- $e_i = y_i - \hat{y}_i$ is the ith residual
- RSS $= \sum_i e_i^2$

- RSS is minimized at *least squares coefficient estimates*

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$

$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$

Evaluating the model

$E(\hat{\beta}_0) = \beta_0$

- Linear regression is unbiased
- Variance of linear regression estimates:

$$\mathrm{SE}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^n (x_i - \overline{x})^2} \right]$$

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \overline{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$

- Estimate σ: $\hat{\sigma}^2 = \frac{RSS}{n-2}$

- The 95% confidence interval for β_1 approximately takes the form

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1)$$

- Hypothesis test:

  H_0: $\beta_1 = 0$
  H_a: $\beta_1 \neq 0$

  - Test statistic $t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}$

Assessing the accuracy of the model

Residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n-2} RSS}$$

n = # data pts

R squared:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

$$TSS = \sum_i (y_i - \overline{y})^2$$

Section 2

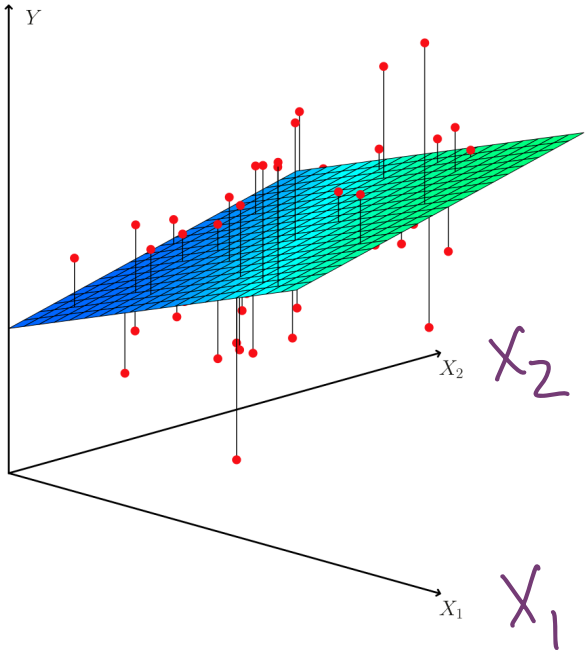
Multiple Linear Regression

Setup

$$\underbrace{Y = \beta_0 + \beta_1 X_1}_{\text{last class}} + \beta_2 X_2 + \cdots \beta_p X_p + \underbrace{\varepsilon}_{\text{noise}}$$

β_j is the average effect on Y of one unit change in X_j

"hyperplane"



Estimation and Prediction

Given estimates $\overbrace{\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \cdots, \hat{\beta}_p}^{p+1 \text{ coefficients}}$, prediction is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$$

Minimize the sum of squares

$$RSS = \sum_i (\underbrace{y_i}_{i\text{th observation}} - \underbrace{\hat{y}_i}_{\text{prediction for } x_i})^2$$

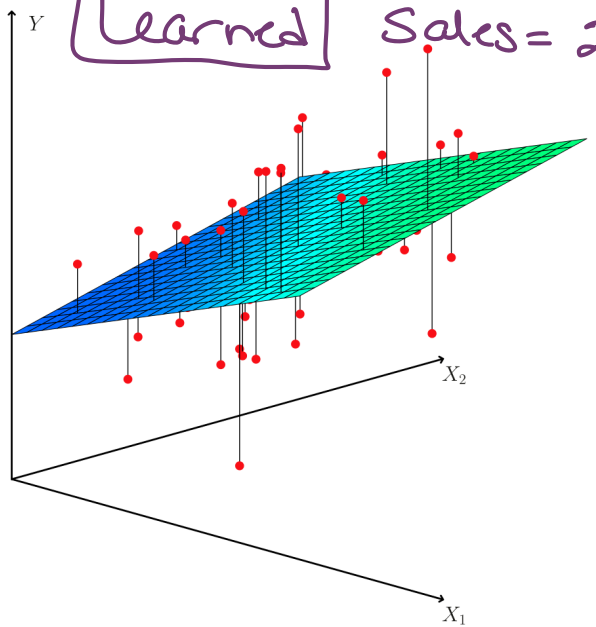
$$= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_1 - \cdots - \hat{\beta}_p x_p)$$

Coefficients are closed form but UGLY

Advertising data set example

$$\texttt{Sales} = \beta_0 + \beta_1 \cdot \underset{X_1}{\texttt{TV}} + \beta_2 \cdot \underset{X_2}{\texttt{radio}} + \beta_3 \cdot \underset{X_3}{\texttt{newspaper}}$$

$\boxed{\text{Learned}}$ Sales = 2.939 + 0.046 TV + 0.189 radio − 0.001 news.



		Coefficient	
	Intercept	2.939	β_0
X_1	TV	0.046	β_1
X_2	radio	0.189	β_2
X_3	newspaper	−0.001	β_3

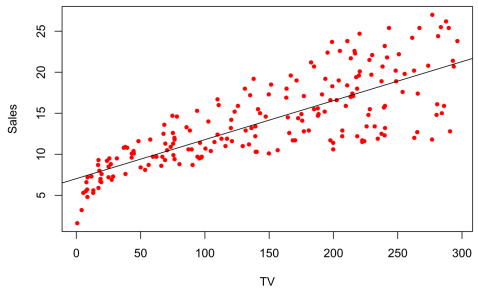
Interpretation of coefficients

$$\texttt{Sales} = \beta_0 + \beta_1 \cdot \texttt{TV} + \beta_2 \cdot \texttt{radio} + \beta_3 \cdot \texttt{newspaper}$$

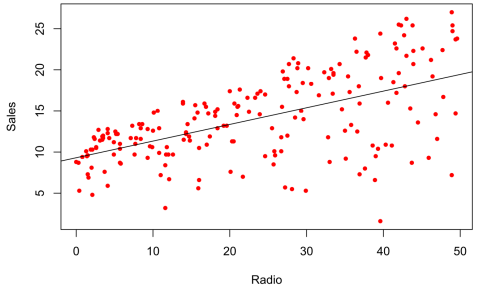
	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

- Look @ radio:
  If newspaper and TV spending are fixed but increase radio spending by \$1,000, then sales are predicted to increase by $(0.189) \cdot 1000 = \$189$
- Look @ newspaper
  Approx 0 coeff, from last class class expect no relationship between newspaper and sales Warning!!!

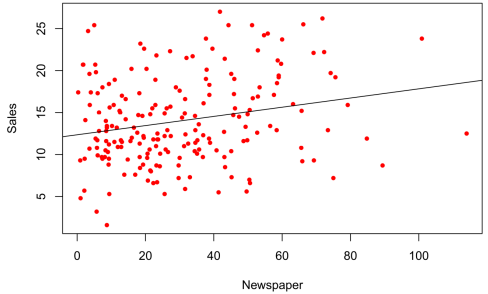
Single regression vs multi-regression



	Coefficient
Intercept	7.0325
TV	0.0475

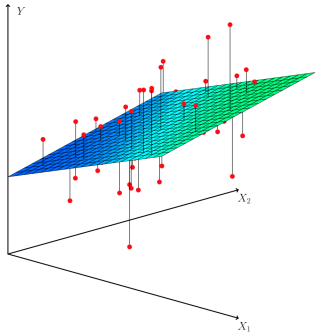


	Coefficient
Intercept	9.312
radio	0.203



	Coefficient
Intercept	12.351
newspaper	0.055

- High correlation b/t radio/news
- Markets w/ lots of radio spending also have lots of news spending



	Coefficient
Intercept	2.939
TV	0.046
radio	0.189
newspaper	−0.001

Correlation matrix

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

In ~~single~~ mult regression, radio gets credit for news spending

Coding group work

Run the section titled "Multiple Linear Regression"

Warning:

Conclusions about low/small coeff should be limited to whether $\beta_i = 0$ instead of "there is a relationship b/t the two"

Section 3

Ch 3.2.2: Questions to ask of your regression

Question 1

Is at least one of the predictors $X_1, \cdots, X_p$ useful in predicting the response?

Q1: Hypothesis test

$H_0: \beta_1 = \beta_2 = \cdots = \beta_p = 0$

$H_a:$ At least one β_j is non-zero

F-statistic:

$$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}$$

# of input variables

# of observations

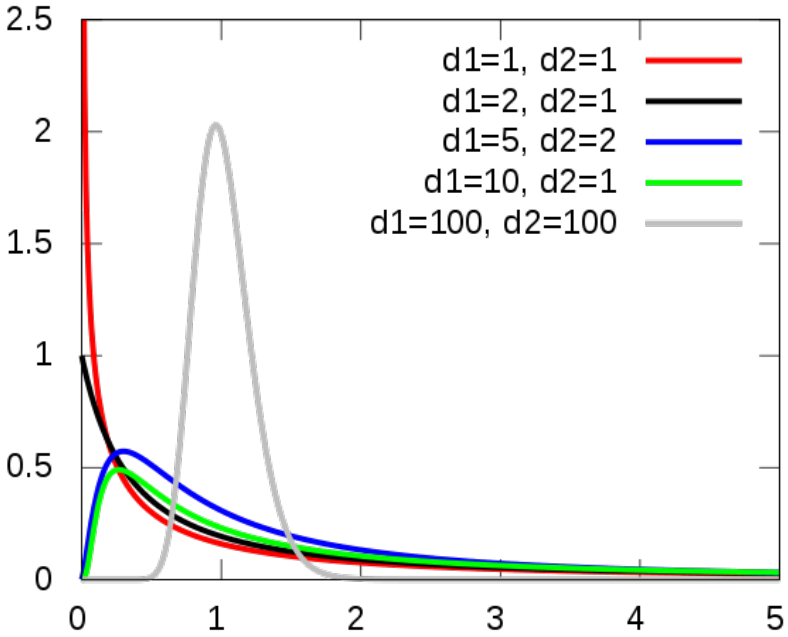
Total Sum of Squares

$\sum (y_i - \hat{y}_i)^2$

$\sum (y_i - \bar{y})^2$

The F-statistic for the hypothesis test

$$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}$$



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- If linear model assumptions are true

$$E\left(\frac{RSS}{n-p-1}\right) = \sigma^2 = Var(\varepsilon)$$

- If in addition H_0 is true

$$E\left(\frac{TSS - RSS}{p}\right) = \sigma^2$$

- So if null hyp is true,

$$F \approx \frac{\sigma^2}{\sigma^2} = 1$$

- On the other hand if alt hyp is true

$$E\left(\frac{TSS - RSS}{p}\right) > \sigma^2$$

So then $F \gg 1$

Do Q1 section in jupyter notebook

Q2

Do all the predictors help to explain Y, or is only a subset of the predictors useful?

Q2: A first idea

Great, you know at least one variable is important, so which is it?....

Bad idea plan A:

Take every possible subset of the variables

Get the least squares fit

Choose the best based on some criteria

ex) TV, radio, news

T R N

{T}

{R}

{N}

{T, R} → $Y \approx \beta_0 + \beta_1 T + \beta_2 R$

{T, N}

{R, N}

{T, R, N}

Do Q2 section in jupyter notebook

Why is this a bad idea?

In gen, have lots of inputs

e.g. $p=40$ input variables

This involves checking ≈ 1 billion models

Better ideas for later (Ch 6 ish)

↳ More intelligent ways of picking

Next time

Lec #	Date			Reading	HW
1	Mon	8/26	Intro / First day stuff / Python Review Pt 1	1	
2	Wed	8/28	What is statistical learning?	2.1	
	Fri	8/30	Class Cancelled (Dr Munch out of town)		
	Mon	9/2	No class - Labor day		
3	Wed	9/4	Assessing Model Accuracy	2.2.1, 2.2.2	
4	Fri	9/6	Linear Regression	3.1	HW #1 Due Sun 9/8
5	Mon	9/9	More Linear Regression	3.1/3.2	
6	Wed	9/11	Even more linear regression	3.2.2	
7	Fri	9/13	Probably more linear regression	3.3	Hw #2 Due Dun 9/15
8	Mon	9/16	Linear regression coding module		
9	Wed	9/18	Intro to classification, Bayes classifier, KNN classifier	2.2.3	
10	Fri	9/20	Logistic Regression	4.1, 4.2, 4.3.1-3	Hw #3 Due Sun 9/22
11	Mon	9/23	Multiple Logistic Regression / Multinomial Logistic Regression /Project day	4.3.4-5	
	Wed	9/25	***Review***		